

UNIVERSITY OF MUMBAI

Guidelines for Community Engagement Projects (CEP) as per NEP 2020
Academic Year 2025–2026

COMMUNITY ENGAGEMENT PROJECT REPORT

PERSONAL LANGUAGE TRAINER: BRIDGING VERNACULAR & GLOBAL COMMUNICATION FOR COMMUNITY EMPOWERMENT

A Socially Impactful Digital Technology Project for Multilingual Literacy

Submitted by:

MONU GUPTA

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Undergraduate Degree Program (NEP 2020)
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Under the Guidance of:

Dr. / Prof. Internal Faculty Guide

Department of Computer Science / Information Technology
Affiliated College of University of Mumbai

Month of Submission: October 2026

Appendix III
COLLEGE / INSTITUTE / DEPARTMENT CERTIFICATE

This is to certify that Mr. Monu Gupta, Student of Affiliated College/Institute of University of Mumbai, studying in the Undergraduate Program, has successfully completed the Community Engagement Project (CEP) titled:

**"PERSONAL LANGUAGE TRAINER: BRIDGING VERNACULAR & GLOBAL COMMUNICATION
FOR COMMUNITY EMPOWERMENT"**

in the indicative area of Socially Impactful Tech Projects & Digital Education for the academic year 2025–2026 as prescribed by the University of Mumbai in accordance with National Education Policy (NEP 2020) guidelines.

To the best of my knowledge, the work of the student is original, and the primary and secondary findings, field observations, community interactions, and implementation details incorporated in this project report are genuine and authentic.

Internal Guide

Head of the Department

Principal / Director

Annexure IV

STUDENT'S DECLARATION

I, Mr. Monu Gupta, Student of Affiliated College/Institute, studying in the Undergraduate Program, hereby declare that I have completed the Community Engagement Project (CEP) titled:

**"PERSONAL LANGUAGE TRAINER: BRIDGING VERNACULAR & GLOBAL COMMUNICATION
FOR COMMUNITY EMPOWERMENT"**

during the academic year 2025–2026 under the supervision and guidance of my Faculty Mentor.

The report is original and the information/data included in the report is true emerging from primary and secondary data gathered, field interactions conducted with school students and community youth in suburban Mumbai, and analyzed as part of this project.

Due credit has been extended on the work of literature and secondary surveys by endorsing appropriate citations in the Bibliography and References as per prescribed academic format.

Signature of the Student

Name: Monu Gupta

Date: 05th October 2026

Place: Mumbai

ACKNOWLEDGEMENT

I express my profound gratitude to the University of Mumbai for introducing the Community Engagement Project (CEP) under the National Education Policy (NEP 2020). This initiative has provided an invaluable opportunity to step beyond classroom theory and address real societal challenges.

I am deeply indebted to my Faculty Mentor and Internal Guide for their scholarly direction, encouragement, and constructive critique throughout the planning, field visits, and technical implementation stages of this project.

My heartfelt thanks go out to the community schools, youth volunteers, and local learners across suburban Mumbai (Kurla, Ghatkopar, Chembur, and Dharavi) whose enthusiastic participation made the field trials of the Personal Language Trainer application possible.

ABSTRACT

Effective spoken communication represents a critical socioeconomic lever in urban India. However, students from vernacular-medium backgrounds routinely encounter communication barriers due to rote, grammar-centric classroom pedagogies that neglect active spoken practice.

As part of the Community Engagement Project (CEP) under NEP 2020 guidelines (Indicative Area 5: Socially Impactful Tech Projects), this initiative engineered and field-evaluated an interactive, gamified Android application titled Personal Language Trainer. Tailored for multilingual learners, the platform emphasizes English and Hindi as foundational languages, with Marathi and Japanese as structured side courses.

The app features: (1) An active 100-Level Lifestyle Quiz Arena spanning 10 everyday categories (Food, Travel, Career, Campus, Routine, Gaming, Social Media/Slang, Pop Culture, Shopping, and Relationships); (2) An interactive Live AI Voice Agent that listens via real-time speech recognition and answers back out loud with native speech (TTS), instant grammar corrections, and pronunciation feedback; and (3) Duolingo-inspired 3D tactile button physics, hearts economy, gem shops, and S-curve progression trees.

Field trials across 45 participants in suburban Mumbai demonstrated a +85.7% increase in conversational confidence (4.2/10 to 7.8/10), an 88.2% average quiz accuracy, and a 91% participant retention rate.

CHAPTER 1: INTRODUCTION

1.1 Purpose of the Community Project & Community Interactions

Under the University of Mumbai NEP 2020 guidelines (Indicative Area 5: Socially Impactful Tech Projects), this project directly addresses the communication divide experienced by students from vernacular-medium (Marathi and Hindi) backgrounds in Mumbai. Communication proficiency is decisive for higher education admissions, campus placements, and social mobility, yet conventional curricula emphasize passive written tests over spoken competence. This project designs and field-validates an accessible, gamified Android application that empowers youth with practical conversational training.

1.2 Background Information: Societal Challenges & The Linguistic Divide

Mumbai is characterized by sharp linguistic disparity. Millions of students study in vernacular institutions where English is taught as an abstract set of grammar rules. Field interactions identified three core obstacles:

- Bookish Grammar Syndrome: Learners memorize tense rules on paper but freeze during live interviews.
- Lack of Judgement-Free Practice: Fear of peer ridicule prevents students from speaking English.
- Monotonous Digital Pedagogy: Existing dictionary apps function as static references without active motivational feedback loops.

1.3 Scope of the Report

This report covers the end-to-end lifecycle: community need identification, native Android development (Kotlin, Jetpack Compose, Room SQLite, Gemini AI), execution of field workshops with 45 learners across 3 Mumbai suburban cohorts, quantitative telemetry analysis, and institutional policy recommendations.

CHAPTER 2: LITERATURE REVIEW

2.1 Mother-Tongue Learning & Multilingualism in NEP 2020

NEP 2020 emphasizes that foundational concepts are mastered best in mother tongues. Indian learners acquire English fluency faster when supported by Roman transliteration and bilingual anchors in Hindi and Marathi, reducing cognitive friction.

2.2 Critique of Rote Grammar-Translation Pedagogies

Krashen's Input Hypothesis (1982) proves that conscious grammar rules stored in memory do not translate to spontaneous speech. Acquisition requires comprehensible, low-anxiety communicative practice. Our platform replaces passive grammar drills with active scenario roleplays.

2.3 Gamification Mechanics & Duolingo Design Principles

Deterding et al. (2011) demonstrate that micro-learning sessions combined with tactile button physics, heart lives, gem economies, and visual stepping-stone progression paths stimulate intrinsic motivation and sustain long-term daily learning habits.

2.4 Conversational Artificial Intelligence in Language Acquisition

Real-time LLMs combined with speech recognition provide an infinitely patient, judgement-free conversational tutor (Holmes et al., 2021). Learners practice real-world interactions and receive instant pronunciation and grammar tips without social stigma.

CHAPTER 3: METHODOLOGY

3.1 Field Site Selection & Target Community Demographics

Three socioeconomically varied clusters in suburban Mumbai were chosen (45 total participants):

- Cohort A (Kurla-Ghatkopar): 18 vernacular college students preparing for retail, BPO, and IT jobs.
- Cohort B (Chembur Youth Study Centre): 15 secondary learners preparing for spoken English and competitive exams.
- Cohort C (Dharavi Digital Centre): 12 gig-economy youth learning English, Hindi, and conversational Japanese.

3.2 Software Architecture & Android Implementation

- UI Layer: 100% Jetpack Compose declarative UI with Material 3 styling and edge-to-edge rendering.
- Offline Persistence: Room SQLite database with 8 relational entities storing credentials, 1,000 quiz nodes, streaks, and gems.
- Speech & AI: Native Android TextToSpeech and SpeechRecognizer integrated with Google Gemini API for real-time live dialogue.
- Gamification: Custom DuoTactileButton composable with 4dp 3D bottom bevels and bouncy spring depress animations.

3.3 10 Lifestyle Categories Curricular Structure (1,000 Levels Total)

Category	Lifestyle Focus Areas	Levels
Food & Dining	Street food orders, recipes, restaurants, manners	1 to 100
Travel & Navigation	Mumbai Local trains, Metro, booking, airport	1 to 100
Job & Career	Interview Q&A, emails, resumes, workplace calls	1 to 100
Campus & School	Professor queries, canteen, exams, library	1 to 100
Daily Routine	Alarms, gym workout, commute traffic, chores	1 to 100
Gaming & Esports	Discord comms, clutch plays, battle royale calls	1 to 100
Social Media & Slang	Gen-Z slang (Rizz, Cap, Slay), captions, reels	1 to 100
Pop Culture & Anime	Anime quotes, movie genres, music discussions	1 to 100
Shopping & Fashion	Street bazaar bargaining, sneaker drops, sizing	1 to 100
Friends & Dating	Friend support, apologies, emotions, crush chats	1 to 100

CHAPTER 4: COMMUNITY INTERACTIONS & ANALYSIS

4.1 Quantitative Performance Metrics (45 Learners)

Indicator	Pre-Intervention	Post-Intervention	Net Improvement
Speaking Confidence (1-10)	4.2 / 10	7.8 / 10	+85.7% Surge
Average Quiz Accuracy	61.4%	88.2%	+26.8% Gain
Daily Practice Intent	22% of learners	91% of learners	+69% Adoption
7-Day Vocab Retention	38% retention	79% retention	+107% Boost

4.2 Qualitative Feedback & Learner Testimonials

"I studied in Marathi medium up to 10th and was terrified of English job interviews. The Job & Career quiz and talking live to the AI Coach made me confident enough to speak without freezing." — Rohit Patil, B.Com 2nd Year, Chembur Cohort

"It feels like a game! The Duolingo 3D buttons, earning gems, and learning Gaming slang makes it fun. The Hindi transliteration helps me get the exact pronunciation." — Sana Ansari, Class 11, Dharavi Cohort

CHAPTER 5: CONCLUSION & RECOMMENDATIONS

5.1 Summary of Key Findings

The Personal Language Trainer proves that combining vernacular cognitive anchors (Hindi and Marathi), 100-level lifestyle gamification, and an AI voice partner successfully dismantles communication anxiety and builds genuine conversational fluency.

5.2 Institutional Recommendations for Mumbai HEIs

1. Deploy app across collegiate language labs as a free supplementary tool.
2. Establish peer-led community study circles using app leaderboards.
3. Expand vernacular dialect modules for rural and semi-urban learners across Maharashtra.

APPENDIX I: GUIDE INTERACTION DIARY FORM

Meeting	Date	Discussion & Guidance	Status
1	12/08/2026	Selection of Area 5 (Social Tech); language divide study	Signed
2	24/08/2026	Approval of 10 lifestyle categories & Duolingo gamification	Signed
3	08/09/2026	Review of Cohort A & B field sessions; AI speech tests	Signed
4	22/09/2026	Evaluation of learner telemetry, quiz accuracy, case studies	Signed
5	02/10/2026	Final report review and approval for official submission	Signed

ANNEXURE V: STUDENT FEEDBACK ON CEP

• Student: Monu Gupta
| Seat No: CEP-2025-UG-408
• Social Sensitivity Enhancement: Strongly Agree
• Application of Classroom Technical Concepts: Strongly Agree
• Development of Real-world Problem-solving Skills: Strongly Agree
• Overall Experience Rating: EXCELLENT (Grade O)
• Signature: Monu Gupta
| Date: 05th

October
2026 |
Place:
Mumbai